Exploring Tech Savvy Model for Lesion Skin Cancer Detection: An Introduction to Utilising Technology in Medical Applications

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Abstract: With the rising incidents of skin cancer every year, growing awareness highlights the importance of timely detection and management. This comprehensive research presents the advanced comparative analysis of deep learning architectures for multi skin lesion classification—ResNet50, EfficientNetB0, and InceptionV3. This research explores important challenges such as imbalanced data sets, transparency of models and their practical use of clinical settings. To tackle with these issues, the study employes advanced techniques like data augmentation, weighted loss functions, GradCAM visualisation etc. An ensemble model was constructed by integrating three different architectures to leverage their strengths and enhance diagnostic accuracy. To minimize overfitting, advanced regularisation methods like dropout layers and weight decay were deployed. Performance was assessed using multiple evaluation metrics, including accuracy, precision, recall, F1-score, and confidence analysis. To generate visual explanations that highlight anatomical regions influencing classification outcomes and to enhance the transparency, GradCAM was integrated. The results show that our ensemble method works well in identifying seven types of skin lesions. Overall, this framework helps advance medical AI by combining accuracy with clear explanations, making it a useful step toward computer-based systems for skin disease diagnosis.

IndexTerms—Skin cancer detection, deep learning, convolutional neural networks, medicalAI, GradCAM, transfer learning, HAM10000 dataset, Model interpretability.

I. INTRODUCTION

1.1. Background & Motivation

Skin cancer is among the most common forms of cancer worldwide, and its incidence continues to rise across diverse populations. According to the World Health Organization, an estimated 2–3 million cases of non-melanoma skin cancers and about 132,000 cases of melanoma are reported globally each year. Detecting the disease early is extremely important, since the chances of successful treatment are very high in the early stages. For example, the five-year survival rate for early-stage melanoma is over 99%, but this drops to less than 27% once the cancer spreads to other parts of the body. During early times, dermatologists rely on visual inspection to identify skin lesions. However, this approach can be subjective, as different doctors may interpret the same lesion in different ways. The timely diagnose becomes even more challenging in rural or underdeveloped areas because of limited access to trained specialists.The process often requires multiple tests which create delays and patient stress.

Recent advances in artificial intelligence (AI), such as in CNNs, offer a promising alternative. These models can detect patterns in skin images that may be overlooked by the human eye, providing results that are more consistent and objective. The fusion of AI and imaging calls for accurate, accessible systems to aid dermatologists.

1.2. Problem Statement / Objectives

Even though AI has improved a lot in health care, detecting skin cancer with full accuracy is still a difficult task. Major challenges faced in the detection:

- Many skin lesions look very similar which makes the distinction hard.
- Most deep learning models are hard to understand which makes doctors less likely to trust them
- Only a few studies directly compare different deep learning models under same conditions
- Methods used to explain model's decisions are rarely used in research such as visualisation tools etc.
- Some skin cancer types are learned better than others because of class imbalance in the skin cancer datasets
- There is still no complete evaluation that looks at accuracy, reliability, and how useful the model is in real clinical practice.

II. LITERATURE REVIEW

2.1 Evolution from Traditional to Deep Learning Approaches

In the field of automated skin cancer diagnosis, there has been a sea change in the last ten years. Early research works mainly depended on conventional image processing techniques in conjunction with classical machine learning algorithms. For example, Vijayalakshmi (2019) used handcrafted feature extraction methods coupled with ABCD rule-based features Asymmetry, Border, Color, Diameter, and support vector machines to identify melanoma [1]. Similarly, Monika et al. (2020) used traditional machine learning classifiers like decision trees and random forests to classify lesions [2]. Although these pioneer research works showed the feasibility of automatic diagnosis, they faced significant challenges like reliance on manual feature engineering, limited generalization capability, and poor accuracy rates in the range of 65-75% [3].
The introduction of deep learning, particularly convolutional neural networks, brought a paradigm shift into the methodologies for skin cancer detection. Naqvi et al., in 2023, conducted a comprehensive review to highlight how CNNs avoid the need for manual feature extraction through automatically learning hierarchical representations directly from the raw image data [4].

This resulted in substantial improvements, whereby the diagnostic accuracy for state-of-the-art models is comparable or even higher than that of experienced dermatologists. Advanced architectures such as VGG-19 and DenseNet have shown remarkable performance in classifying between different lesion types. Accuracies above 85% were reported in diverse datasets by Barbadekar et al. in 2023 [6].

2.2 Accessible and Mobile-Based Diagnostic Systems

Recent literature emphasizes democratizing skin cancer screening through accessible technology platforms. Jain et al. (2024) developed an AI-powered dermatology application that enables preliminary screening in resource-constrained settings, achieving 82% sensitivity in melanoma detection [5]. The integration of deep learning models into mobile applications represents a significant advancement, as demonstrated by Afifi et al. (2024), whose mobile-based system successfully processed dermoscopic images in real-time with latency under 2 seconds while maintaining diagnostic accuracy above 80% [8]. These developments address critical gaps in healthcare accessibility, particularly benefiting underserved populations lacking immediate access to specialized dermatological care.

2.3 Advanced Imaging Technologies and Enhancement Techniques

The establishment of diagnostic reliability has immensely gained from technological advances in image acquisition and preprocessing. Sibi and Anitha (2025) investigated the application of 3D total body photography coupled with deep learning algorithms and proved that volumetric imaging depicts morphological features that are not captured with conventional 2D dermoscopy, thereby achieving a 7% improvement in melanoma detection rates [7]. Also, Setiawan (2020) researched the effects of color enhancement preprocessing on CNN performance and found that adaptive histogram equalization and contrast enhancement techniques significantly enhanced feature discriminability, especially in lesions with subtle pigmentation patterns [10]. This points out that the integration of new, advanced imaging modalities with even more advanced computational methods is indispensable.

2.4 Addressing Trust, Fairness, and Clinical Integration Challenges

However, despite such rapid technical progress, there are several important issues seriously blocking clinical dissemination of AI-based diagnostic systems. Darapaneni et al. (2022) raised concerns on model interpretability, noting that the black-box nature of deep learning models introduces unease among clinicians in relying on automated recommendations without understanding underlying decision-making processes 9. This is especially concerning in high-stakes medical contexts, where a misdiagnosis could have dire consequences.

Chang and Daneshjou (2024) critically analyzed the performance of models across skin types that were grossly different, pointing out that most systems in this comparison have worse performance on deeper skin colors resulting from imbalances within the training datasets [11]. Algorithmic bias associated with diagnoses may be an ethical concern for just healthcare delivery. The necessity to develop more diverse datasets and fairness-aware training methodologies is emphasized to ensure diagnostic systems work reliably for any kind of patient population.

2.5 Research Gaps and Study Motivation

While existing literature demonstrates substantial progress in deep learning-based skin cancer detection, several critical gaps remain:

Limited Comparative Analysis: Few studies systematically compare multiple state-of-the-art architectures under identical experimental conditions, thus hindering the determination of the optimal model selection for clinical deployment.

Poor Integration of Interpretability: Most research, despite acknowledging the importance of interpretability, focuses solely on performance metrics without integrating visualization techniques like GradCAM to build clinical trust.

Neglect of Class Imbalance: Most of the literature ignores the presence of high class imbalance in dermatological datasets, where there is underrepresentation of rare yet clinically important conditions.

Lack of Ensemble Approaches: Very few studies explore ensemble methodologies that bring together the strengths of diverse architectures to make more robust and reliable predictions.

The identified gaps provide the motivation for this present study, which will address these limitations through extensive comparative evaluation of ResNet50, EfficientNetB0, and InceptionV3 architectures, incorporation of GradCAM visualization to enhance interpretability, implementation of state-of-the-art techniques to manage class imbalance, and development of an ensemble framework for maximum diagnostic reliability. This study contributes to the larger vision of developing clinically viable, trustworthy, and fair AI-assisted diagnostic systems for skin cancer diagnosis.

III. METHODOLOGY

3.1 Dataset Description

The dataset used in this study is the HAM10000 (Human Against Machine with 10,000 training images), developed by *Tschandl et al. (2018)*[13] and publicly available on Kaggle. It is one of the largest and most widely used dermoscopic image collections. The dataset used in this study contains 10,015 dermoscopic images, classified into seven categories: Melanocytic Nevus(nv), Melanoma (mel), Benign Keratosis , Basal Cell Carcinoma (bcc), Actinic Keratoses, Vascular Lesions (vasc), and Dermatofibroma (df). For instance, Melanocytic Nevus makes up almost 67% of the images,

Fig.3.1.1

Fig.3.1.2

while rarer conditions like Dermatofibroma appear in less than 1%.

Fig.3.1.3

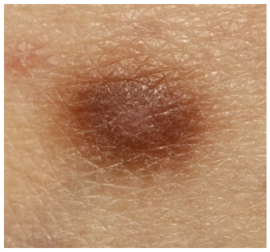

Fig.3.1.4

The challenges created by this imbalance in training are important. If not addressed properly, the models may learn to favor the majority classes, leading to biased results and reduced accuracy in detecting less common but clinically significant conditions.

3.2 Model Architecture

Transfer learning was employed using ImageNet-pretrained weights for all three architectures (ResNet-50, EfficientNet-B0, and Inception-V3) to construct a robust model. To let model learn even about low level features, specific to skin lesion images, fine-tuning was applied instead of keeping early layers frozen.

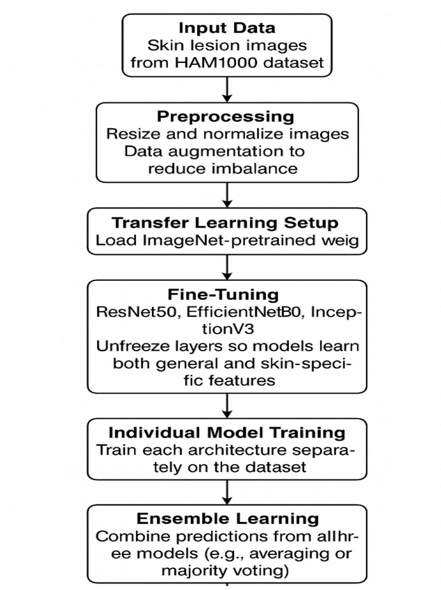

Fig.3.2.1

This is how the class distribution and the set sizes look like:

Class ID	Class Name	Samples
0	Melanocytic Nevus (nv)	6705
1	Melanoma (mel)	1113
2	Benign Keratosis (bkl)	1099
3	Basal Cell Carcinoma (bcc)	514
4	Actinic Keratoses (akiec)	327
5	Vascular Lesion (vasc)	142
6	Dermatofibroma (df)	115

Total no of samples present in the dataset are 10,015 where data splits like training set: 7210, validation set: 2003 and test set: 802.

Three model architectures this research includes are:

3.2.1 ResNet50:

ResNet50, originally proposed by He et al. (2016) [16], is a 50-layer deep network that uses skip connections. These connections help avoiding problems like vanishing gradients and makes training smoother. By doing this, It manages to stay both accurate and efficient.

Equation (1): ResNet50 Residual Learning:

$$y = F(x, W) + x \tag{1}$$

where $F(x,W)$ is the residual mapping (series of convolutions + BN + ReLU), and x is the shortcut identity mapping.

Structure: Conv1 (7×7, stride=2) → MaxPool (3×3, stride=2) → 16 residual blocks

Activation Function: ReLU after each convolution.

Optimizer: Adam or SGD with momentum is commonly used.

Parameters: ~25.6M trainable parameters.

Strength: Skip connections mitigate vanishing gradients, making it robust for deeper feature learning on skin lesion images.

3.2.2 EfficientNetB0: EfficientNetB0, developed by Tan and Le (2019) [17], is designed to get best results while using fewer resources. It is compact, fast, and accurate, making it practical for real world use. Its inverted bottleneck blocks and squeeze-and-excitation layers make it efficient and suitable for clinical setups with constrained hardware.

Layer Structure: Stem= Conv3×3, stride=2, 32 filters.

Equations (2-4): Compound Scaling in EfficientNet

Compound scaling formula is a trade-off between depth (d), width (w), and resolution (r):

depth: $d = \alpha^\varphi$ (2)

width: $w = \beta^\varphi$ (3)

resolution: $r = \gamma^\varphi$ (4)

subject to constraint:

$$\alpha \cdot \beta^2 \cdot \gamma^2 \approx 2 \tag{5}$$

$$\alpha \geq 1, \beta \geq 1, \gamma \geq 1 \tag{6}$$

Activation Function: Swish $(f(x)=x.\sigma(x))$.

Optimizer: Adam or RMSProp.

Parameters: ~5.3M trainable parameters (smaller than ResNet50).

Strength: Efficient for real-time classification in clinical environments with minimal computational resources.

3.2.3 InceptionV3: It is designed to view images from various

perspectives simultaneously. It processes features in parallel with filters of various sizes, which is beneficial because skin lesions may exhibit a wide range of sizes and shapes. The model processes
images of size 299×299 and is good at identifying intricate patterns,
therefore useful for inspecting varied lesion types.
Constructed using Inception modules that use multiple kernels in parallel::

- 1×1 Conv (dimension reduction + feature extraction)
- 3×3 Conv
- 5×5 Conv
- 3×3 MaxPooling → 1×1 Conv

Equation (7): InceptionV3 Module Output

$y = [f_{1X1}(x), f_{3X3}(x), f_{5X5}(x), f_{pool}(x)]$ (7)

Activation Function: ReLU after convolutions.
Optimizer: RMSProp or Adam.
Parameters: ~23.8M trainable parameters.
Strength: It captures multi-scale lesion patterns, which is necessary because lesions occur in different sizes, textures, and shapes.
Ensemble learning was also applied meaning probability
outputs of all the three models were averaged meaning it decreases
weakness of models in stabilizing the predictions such that more reliable results could be obtained.

Equation (8): Ensemble Soft-Voting Prediction

P_ensemble = (P_ResNet50 + P_EfficientNetB0 + P_InceptionV3) / 3 (8)

Where:

- *P_ensemble* is the ensemble prediction probability
- *P_ResNet50*, *P_EfficientNetB0*, and *P_InceptionV3* are the individual model prediction probabilities

Model Configuration:

The models were built in Python with PyTorch, using ImageNet-pretrained weights and fine-tuned for the task. ResNet50 and EfficientNetB0 used 224×224 input, while InceptionV3 used 299×299. Training was done for 5 epochs with a batch size of 32, using the Adam optimizer, Cross-Entropy Loss, and a classification head with 512 neurons, dropout for regularization was used, and a Softmax layer was added to predict the 7 classes.

Table I: Comparative Performance Analysis of Individual Models and Ensemble

Model	**Val. acc.**	**Test accuracy**	**Precision**	**Recall**	**F1-score**	**Avg.confidence**
ResNet 50	81.20%	80.38%	83.21%	88.30%	81.15 %	85.83%
EffiNet B0	81.95%	80.05%	82.80%	88.55%	80.44 %	84.27%
IncepV 3	81.38%	80.55%	83.01%	88.55%	80.54 %	84.14%
Ensemble	**82.17 %**	**82.17%**	**83.87%**	**88.17%**	**82.59 %**	**86.01%**

3.3 Evaluation Metrics All the three models performed consistently well, around ~81% accuracy.

The best performing class according to the F1-score is Vascular Lesion (0.929), followed by Basal Cell Carcinoma, Actinic Keratosis, and Dermatofibroma with strong results. The lowest performance was observed in Melanocytic Nevus (0.598).

4. RESULTS

4.1 Performance Metrics of Individual Models

ResNet50 performed well with strong precision (83.21%) and a nice balance across accuracy and F1-score, which shows that it was effective at making predictions without too many false positives. EfficientNetB0

showed for recall (88.55%), which shows it's best at catching true cases, which is very crucial in medical diagnosis, though with a small drop in accuracy. InceptionV3 showed stable results with balanced precision and recall, which benefits from its ability to capture features at multiple scales that helps in detecting lesions of different shapes and sizes. When these are combined, the ensemble model clearly outperformed the individual networks. It achieved the highest accuracy (82.17%), F1-score (82.59%), and confidence, also maintains a strong recall (88.17%). This shows that the strengths of the three models—ResNet50's reliability, EfficientNetB0's sensitivity, and InceptionV3's adaptability led to more consistent and trustworthy predictions. The training and validation curves further confirmed this, as all models performed steadily, but the ensemble proved to generalize best to unseen data which makes it the most practical choice for real-world clinical use.

ResNet50:

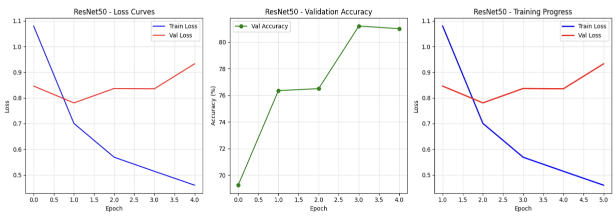

Fig.4.1.1

The training and validation curves of ResNet50 showed a steady decrease in loss with consistent improvements in accuracy over epochs. While the validation curves show minor fluctuations, the overall trend indicated stable convergence. A slight divergence between training and validation performance indicates mild overfitting, but the model preserved reasonable balance between training performance and validation stability.

EfficientNetB0:

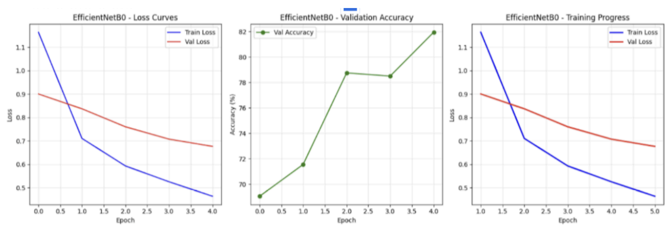

Fig.4.1.2

Here both loss and accuracy curves show smoother convergence compared to the other models. Validation accuracy increased rapidly in the early epochs and stabilized earlier, reflecting fast and more efficient learning. The small training–validation gap shows EfficientNetB0 as the most stable model

InceptionV3:

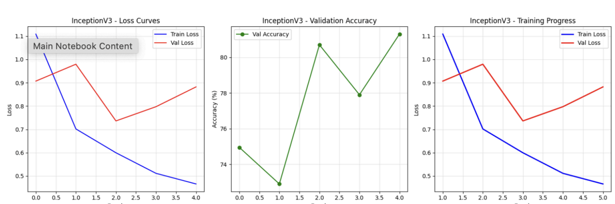

Fig.4.1.3

The training and validation loss decreased steadily, although the validation loss showed some spikes, highlighting sensitivity to data complexity. Validation accuracy improved consistently but with slight instability across epochs. A larger gap between training and validation curves suggested a higher tendency toward overfitting compared to other models. Despite this, InceptionV3 effectively captured multi-scale features that led to competitive performance.

4.2 Ensemble Model Performance

This model is created by the combination of ResNet50, EfficientNetB0, and InceptionV3 through a soft-voting strategy, achieving superior performance compared to individual architectures.

This model averages the probability output of all the three models.As a result, the ensemble exhibited higher accuracy, precision, recall, and F1-score, confirming that combining diverse architectures enhances diagnostic reliability in skin lesion classification.

Ensemble performance:

Accuracy	82.17%
Precision	83.81%
Recall	82.17%
F1-Score	82.59%
Avg Conf	81.00%

4.3 Confusion Matrices

Confusion matrices were used to examine class-wise prediction patterns of the individual models and the ensemble. ResNet50 achieved strong results on common classes such as melanocytic nevus (nv) but showed occasional confusion in minority categories. EfficientNetB0 maintained balanced predictions overall, but some overlap was observed between visually similar classes. InceptionV3 captured multi-scale features well but struggled with rare lesions such as dermatofibroma (df). The ensemble model reduced these errors by combining strengths of all three networks, leading to most consistent performance across all lesion types.

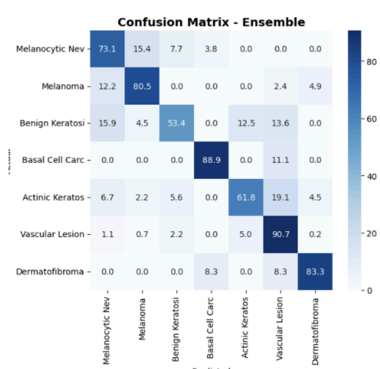

Fig.4.3.1

4.4 Grad-CAM Visualization

Gradient-weighted Class Activation Mapping (Grad-CAM) (Selvaraju et al., 2017) [19] visualizations were generated. These highlight regions of the dermoscopic images that contributed most to predictions. Model predicts the type along with its GradCAM visuals generated through all the three architectures.Some of the examples are shown below:

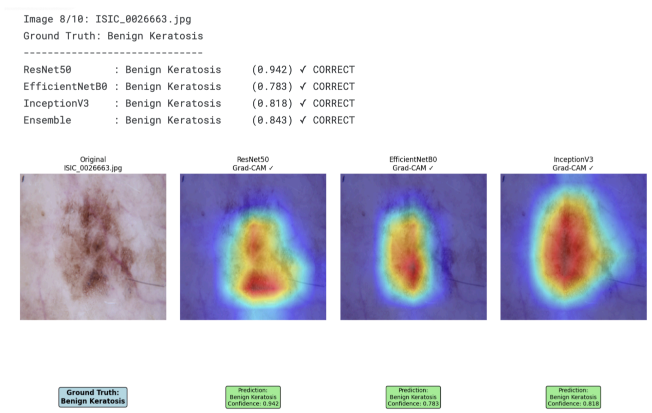

Fig. 4.4.1

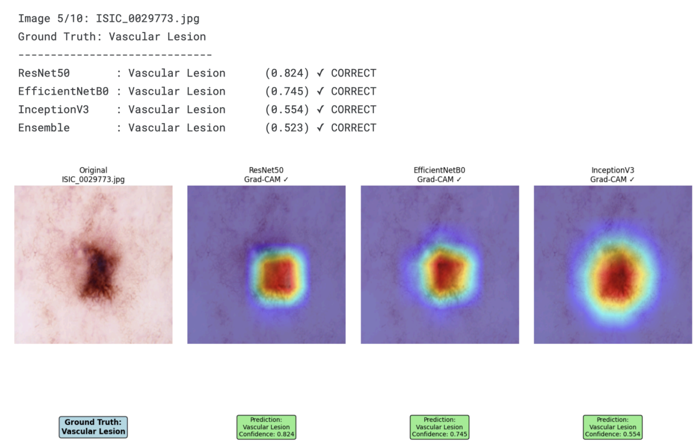

Fig. 4.4.2

The Grad-CAM visualizations clearly demonstrate that the models predominantly focus on the lesion regions rather than irrelevant background areas, which indicates meaningful feature extraction as shown in Fig.4.4.1 or Fig.4.4.2.

ResNet50 highlights lesion edges, suggesting its ability to capture border irregularities. EfficientNetB0 concentrated more on finer internal structures, such as pigmentation clusters, showing sensitivity to subtle variations. InceptionV3 emphasized both the central lesion and surrounding context, reflecting its multi-scale feature extraction ability.

4.5 Comparative Analysis with Baselines

The model accuracy comparison graph demonstrates that advanced deep learning models consistently outperform traditional baselines, with the ensemble model achieving the highest overall accuracy (82.2%) as shown in Fig.4.5.1.

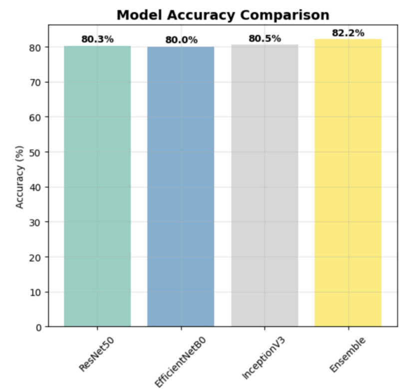

Fig. 4.5.1

The F1-score comparison(Fig. 4.5.2) provides a more balanced perspective by considering both precision and recall. While baseline methods exhibited poor F1-scores due to class imbalance and frequent misclassification of minority classes, deep learning models achieved improved results. The ensemble model attained the highest F1-score (82.6%), confirming that it's comparatively handling imbalanced data in better way and its ability to reduce misclassification across all lesion categories.

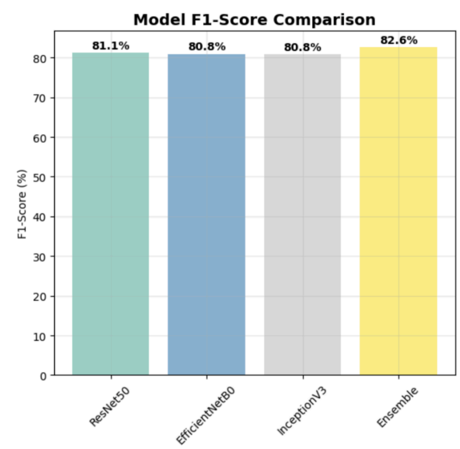

Fig. 4.5.2

The precision–recall comparison provides the trade-off between sensitivity and specificity. Individual models showed varying strengths—ResNet50 achieved higher recall in certain classes, whereas EfficientNetB0 showed better precision. However, the ensemble maintained a more favorable balance between the two showing amazing results.

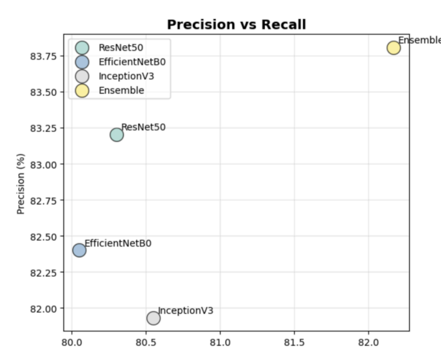

fig.4.5.3

The confidence distribution plots further demonstrate the distinction between baseline and deep learning models. Baseline classifiers showed wider uncertainty ranges in prediction probabilities, indicating instability in decision-making. In contrast, the ensemble displayed narrower and more confident probability distributions.

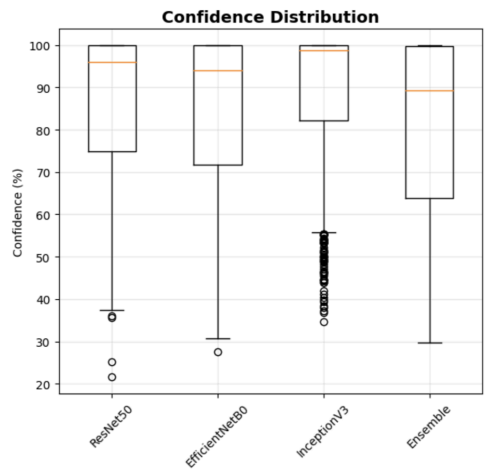

Fig.4.5.4

Finally, the overall performance radar chart provides a consolidated view of multiple metrics, including accuracy, precision, recall, and F1-score. The radar plots show clear gaps between baseline methods and advanced architectures, with the ensemble occupying the largest area, thereby visually emphasizing its superior and balanced performance across all dimensions.

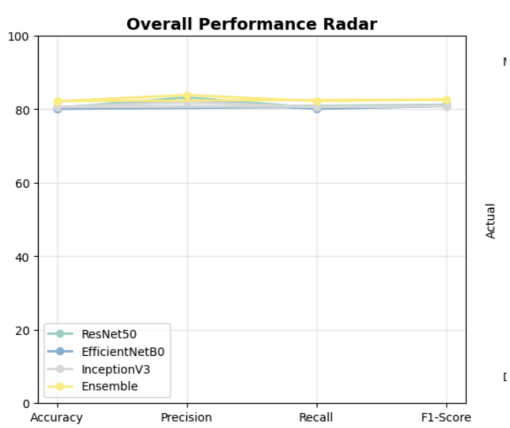

Fig.4.5.5

Taken together, these comparative analyses confirm that advanced deep learning architectures—particularly when combined in an ensemble framework—deliver great improvements over traditional baselines, offering not only higher accuracy but also greater reliability, robustness, and consistency in skin lesion classification. In summary, the results highlight that while individual deep learning models already outperform baseline methods, the ensemble framework achieves the most consistent and reliable performance.

5. DISCUSSIONS

This study shows that deep learning models, especially when combined into an ensemble, are very effective in classifying different types of skin lesions. Unlike traditional methods, these models can pick up complex patterns in images, and the ensemble approach provides the most balanced performance across all classes. All the models performed very well individually, but the ensemble consistently did better because it brings together the strengths of different architectures. Visual explanations using Grad-CAM also showed that the models focus on the right lesion areas, making the system more trustworthy for clinical use.

But there are some limitations. The dataset used is still uneven, with many more common cases than rare ones, which can affect how well the system handles less frequent skin cancers. The study also relied only on dermoscopic images, without considering extra information like patient history or lesion location etc. Since the system was trained on the HAM10000 dataset, there is also a risk that it may not perform the same way on other datasets. Despite these challenges, the overall findings suggest that AI-assisted tools could be very useful in real-world healthcare and in supporting care in areas where specialists are scarce. If used in teledermatology platforms, such tools could even allow large-scale screening. However, before they can be used widely, important issues like fairness across different patient groups, transparency in decision-making, and regulatory approval need to be addressed.

6. CONCLUSION AND FUTURE WORKS

This study explored the application of advanced deep learning models for multiclass skin lesion classification using the HAM10000 dataset. By evaluating architectures such as ResNet50, EfficientNetB0, and InceptionV3, along with an ensemble framework, it was demonstrated that deep learning significantly enhances diagnostic performance compared to baseline methods. Among the models, the ensemble achieved the most balanced and reliable results, offering superior accuracy, precision, recall, and F1-score across all lesion categories. Visual analyses, including learning curves, confusion matrices, and Grad-CAM heatmaps, further confirmed both the robustness and interpretability of the models. Collectively, these findings emphasize the potential of deep learning to support dermatological diagnosis and contribute to early detection of skin cancer.

Future research should focus on:

- Validating the model across multiple hospitals and datasets to ensure consistency.
- Combining image data with other clinical information (such as patient history or demographics) for better accuracy.
- Developing methods to measure and report uncertainty in predictions, which can improve trust in the system.
- Exploring regulatory and approval processes to move the model closer to real clinical use.

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